MIL-L-22620D(AS)
3 February 1976
Superseding
MIL-L-22620C(WEP)
25 February 1971

MILITARY SPECIFICATION

LAUNCHER, GUIDED MISSILE, LAU-7 SERIES

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification establishes the requirements for the manufacture, inspection and testing of the subject guided missile launchers. The test limits in this specification do not necessarily define maximum capabilities. The guided missile launchers are capable of retaining and releasing a missile or a target rocket.

1.2 **Classification.** Guided missile launchers, hereinafter called launchers, covered by this specification shall be of the following types:

Type I LAU-7/A Series Guided Missile Launcher (Navy type)

Type II LAU-7A/A Series Guided Missile Launcher (Air Force type)

2. APPLICABLE DOCUMENTS

2.1 **Specifications, standards and drawings.** The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-640	Box, Fiberboard, Corregated, Triple-wall

FSC-1055

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Military

MIL-P-116	Preservation, Methods of
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-T-7743	Testing, Store Suspension Equipment; General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-V-23254	Valve, Control
MIL-P-81001	Power Supply, PP-2581/A
MIL-R-81202	Receiver Assembly, Nitrogen, for LAU-7/A-3 Launcher

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MS17984	Pin, Quick Release, Positive Locking Single Acting, Button Handle

DRAWINGS

Naval Air Systems Command

DL58A164H791	Data List, Guided Missile Launcher, LAU-7/A-3 (Navy type)
DL64A27H1	Data List, Guided Missile Launcher, LAU-7A/A (Air Force type)
DL60A89D114	Data List, PP-2581/A Power Supply

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 **Precedence of documents.** When the requirements of the contract, this specification or applicable subsidiary specifications are in conflict, the following precedence shall apply:

2.2.1 **Contract.** The contract shall have precedence over all referenced documents.

2.2.2 **Specification.** This specification shall have precedence over all applicable subsidiary documents.

2.2.3 **Referenced specifications.** Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein.

3. REQUIREMENTS

3.1 **First article.** The launchers furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein (see 4.4).

3.2 **Selection of specifications and standards.** Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 **Materials.** Materials shall be as specified on the drawings listed in the data list for the launcher being procured (see 2.1).

3.4 **Interchangeability.** All parts having the same part number shall be functionally and dimensionally interchangeable. Except as otherwise specified, component parts, having the same part number, shall assemble without selection or fitting beyond the necessary matching adjustments shown on the applicable drawings. Applicable provisions of MIL-I-8500 shall apply.

3.5 **Identification of parts.** All parts shall be marked as specified on the drawings listed in the data list for the launcher being procured (see 2.1) or in accordance with MIL-STD-130 if not specified on the drawings.

3.6 **Launcher.**

3.6.1 **Construction.** The construction of the launcher shall be in accordance with drawings listed in the data list for the launcher being procured (see 2.1) and as specified herein.

3.6.2 **Performance.** The launcher, when constructed in accordance with the drawings listed in the data list for the launcher being procured (see 2.1), shall be capable of retaining and releasing missiles specified in 4. The launcher shall meet the requirements of this specification as listed in 4. (See Tables II and III.)

3.6.3 **Power requirements.** The launcher shall perform satisfactorily under all applicable conditions specified herein from a 28 VDC and 115 VAC, 400 Hz power source with characteristics and limits as defined in MIL-STD-704, except that the launcher shall operate over a range of 17 to 29 VDC and 105 to 122 VAC, 380 to 420 Hz, measured at the launcher's input connector (J1).

3.6.4 **Nameplates.** Nameplates for equipment identification shall be in accordance with the drawings listed in the data list for the launcher being procured (see 2.1).

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3.6.5 Nitrogen supply gas. The launcher, while in storage or shipment, shall have the nitrogen receiver pressurized to 225 ± 25 psig (pounds per square inch gage) at room temperature. The gas used in the gas system shall be in accordance with BB-N-411, Type I, Class I, Grade A, except that the maximum contaminant level shall not exceed the levels shown in Table I.

TABLE I

NITROGEN SUPPLY GAS - ACCEPTABLE CONTAMINANT LEVEL

CONTAMINANT	MAXIMUM CONTAMINANT LEVEL
Water (H_2O)	2 parts per million (by volume)
Carbon Dioxide	2 parts per million (by volume)
Freon	3 parts per million (by volume)
Total Hydrocarbons (Including Methane and Acetylene)	10 parts per million (by volume)
Methane	6 parts per million (by volume)
Acetylene	1 part per million (by volume)
Hydrocarbons (others)	3 parts per million (by volume)
Foreign Particles	Maximum of 100 particles, 5.0 to 10.0 microns per cubic foot of flow. No particle larger than 10.0 microns allowed.

3.7 Launcher Components.

3.7.1 Nitrogen receiver. This requirement applies to Type I launchers only. The nitrogen receiver assembly shall be in accordance with MIL-R-81202 and shall be qualified (including all of its components) prior to assembly within the launcher.

3.7.2 Control valve. The control valve shall be in accordance with MIL-V-23254 and shall be qualified prior to assembly within the launcher, unless otherwise authorized by the procuring activity.

3.7.3 Power supply. The power supply PP-2581/A shall be in accordance with MIL-P-81001.

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3.8 **Workmanship.** The workmanship displayed in fabrication and assembly of the launchers shall be such as to assure, within design limitations, ability of the launchers to meet their performance requirements under all applicable environmental conditions specified herein. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assemblies, welding, brazing and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in the approved first article sample, subject to any qualification stated in the government's notice of approval, shall determine the requirements of the remainder of the contract relative to workmanship.

4. QUALITY ASSURANCE PROVISIONS (see 6.4)

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, if approved by the government. The government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to insure that the supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** The inspection of all types of launchers shall be classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 **Inspection conditions.** Unless otherwise specified, all inspections shall be performed under the following conditions and tolerances.

a. Room temperature. Room ambient of 25 ± 10°C (77 ± 18°F) indicated.

b. Test temperature. All parts of the test item shall be stabilized at the specified temperature ± 2.8°C (± 5°F) indicated prior to conducting any operational tests. Unless otherwise specified, temperature stabilization will have been attained when the indicated temperature of the surface of the largest mass of the test item does not change more than ± 2.8°C (± 5°F) in a period of one hour.

c. Humidity. Room ambient to 95 percent relative. When humidity testing is required, the tolerance shall be ± 5 percent relative humidity.

d. Altitude. Unless otherwise specified, testing shall be performed at normal ground elevation. When altitude testing is required, the tolerance shall be ± 5 percent in feet.

e. Test gas. The test gas shall be nitrogen in accordance with 3.6.5.

f. Test missiles. Unless otherwise specified in the applicable paragraph of this specification, the test missile shall be either a Sidewinder 1A or a Sidewinder 1C dummy missile. As far as practicable, dummy (inert) missiles rather than simulated (fabricated) ones shall be used in the tests (see 6.6). The dummy missile specifications shall be based on the following:

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1. Sidewinder 1A (Air Force type)	AIM-9E	170 Pounds
	AIM-9J	170 Pounds
2. Sidewinder 1C (Navy type)	AIM-9D	198 Pounds
	AIM-9G	198 Pounds
	AIM-9H	198 Pounds

Testing will be conducted using the minimum and maximum forward hanger conditions as specified in Figure 1 for the Sidewinder 1A or Figure 2 for the Sidewinder 1C.

g. Power supplies. In the event the contract requires the contractor to provide a power supply with each launcher, this power supply shall be used in the performance of the testing specified herein. The operation shall conform to the power input and load conditions of MIL-P-81001. If no power supply is required, a dummy supply simulating the envelope dimensions, weight, C.G. location and electrical connections of a PP-2581/A power supply shall be used in the performance of testing.

4.4 First article inspection. First article inspection shall be performed on sample units which have been produced with equipment, procedures and materials that will be used for the production of all the launchers designated in the contract or purchase order. First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the government to other contracts or purchase orders.

4.4.1 Launcher samples. First article launcher samples shall consist of the first four (4) launchers manufactured on contract or purchase order. The launchers shall be permanently labeled for ease of identification and shall be produced with equipment and procedures that will be used for the production of all the launchers designated in the contract or purchase order. The launchers shall be tested as indicated in Table II. The procuring activity reserves the right to delete tests and alter the test sequence of Table II. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including statements of findings and certificates of conformance for materials and tests A, B, C, E, F, and H of Table II. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

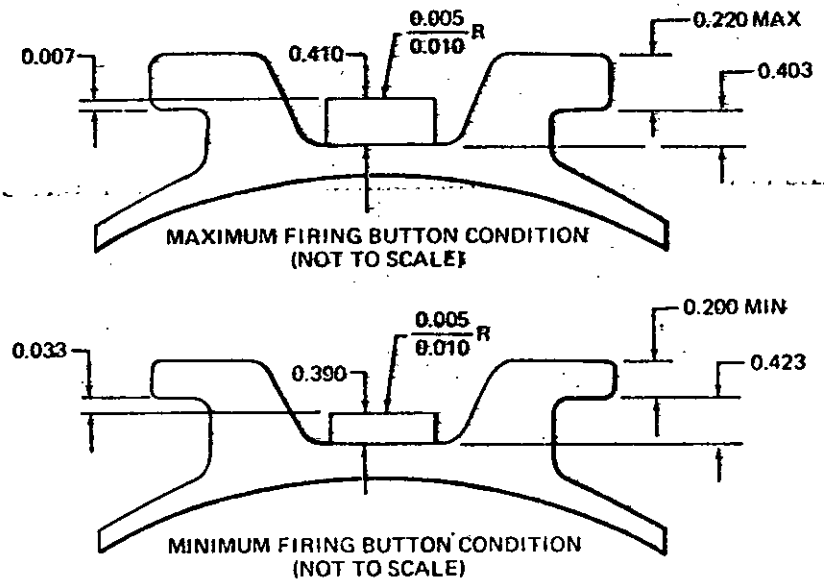
4.5 Quality conformance inspection. Quality conformance inspections consist of those tests conducted on launchers submitted for acceptance under the contract or purchase order. Acceptance of launchers shall be on an inspection lot basis. Quality conformance inspection shall consist of the following tests:

a. Individual tests (see Table III, Group 1)

b. Sampling tests (see Table III, Groups 1 and 2)

4.5.1 Individual tests. Individual tests are those tests conducted by the contractor on each launcher submitted for acceptance under the contract or purchase order. Individual tests for launchers are shown on Table III, Group 1. Failure to pass these tests shall be cause for rejection.

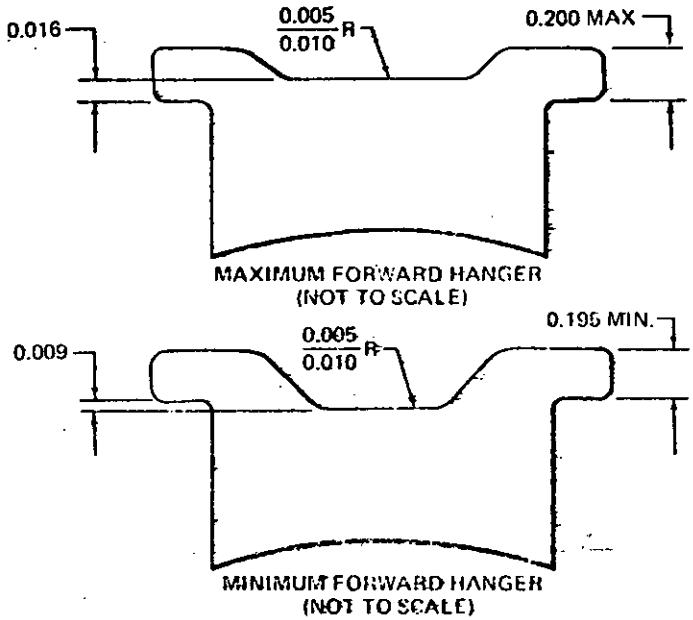
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NOTE: THE LEADING EDGE OF THE FIRING BUTTON SHALL HAVE A RADIUS OF 0.005 TO 0.010 AS INDICATED

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Figure 1. Sidewinder 1A(AIM-9E, 9J) Maximum and Minimum Forward Hanger (Firing Button) Conditions



NOTE: THE LEADING EDGE OF THE CENTER SECTION SHALL HAVE A RADIUS OF 0.005 TO 0.010 AS INDICATED

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Figure 2. Sidewinder 1C(AIM-9D, 9G, 9H) Maximum and Minimum Forward Hanger (Firing Button) Conditions

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TABLE II

FIRST ARTICLE INSPECTIONS - SCHEDULE AND SEQUENCE[1]

TEST	TITLE	TEST PARA.	SAMPLE			
			A	B	C	D
A	Inspection, Launcher	4.6.1	1	1	1	1
B	Nitrogen Gas System[2]	4.6.2	2	2	2	2
C	Functional, Electrical	4.6.3.2	3	3	3	3
D	Electromagnetic Compatibility[3]	4.6.12	4	X	X	X
E	Dielectric	4.6.5.5	5	4	4	4
F	Insulation Resistance (Room Temp.)	4.6.5.6	6	5	5	5
G	Insulation Resistance (High Temp.)	4.6.5.6	7	6	6	6
H	Functional, Mchanical	4.6.3.1	8	7	7	7
I	High Temperature	4.6.8.1	9	X	8	X
J	Low Temperature Leakage[2]	4.6.2.2	10	X	9	X
K	Low Temperature	4.6.8.2	11	X	10	X
L	Temperature Shock	4.6.8.3	12	X	11	X
M	Elevated Temperature	4.6.8.4	13	X	12	X
N	Humidity	4.6.8.11	14	X	13	X
O	Vibration	4.6.8.9	15	X	14	X
P	Shock	4.6.8.10	16	X	15	X
Q	Dust	4.6.8.5	X	8	X	8
R	Frozen Sand	4.6.8.6	X	9	X	9
S	Salt Fog	4.6.8.7	X	10	X	10
T	Altitude	4.6.8.8	X	11	X	11
U	Static Load	4.6.9	X	12	X	12
V	Umbilical Hook Support Assembly	4.6.10	17	13	X	X
W	Firing	4.6.11	18	14	X	X

[1] Test sequence is denoted by the numbers in the sample columns. An X denotes test not required.

[2] Low temperature leakage shall be performed while conducting low temperature tests.

[3] The electromagnetic compatibility test shall not be conducted by the contractor. The test shall be conducted by the government (procuring activity) approved test laboratory on one unit only.

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TABLE III

QUALITY CONFORMANCE INSPECTIONS - SCHEDULE AND SEQUENCE[1]

TESTS	TEST PARA.	SAMPLE			
GROUP 1		ALL SAMPLES			
Inspection, Launcher	4.6.1	1			
Nitrogen Gas System[2]	4.6.2	2			
Functional, Electrical	4.6.3.2	3			
Electromagnetic Compatibility[3]	4.6.12	4			
Dielectric	4.6.5.5	5			
Insulation Resistance (Room Temp.)	4.6.5.6	6			
Insulation Resistance (High Temp.)[2]	4.6.5.6	7			
Functional, Mechanical	4.6.3.1	8			
GROUP 2[4]		A	B	C	D
High Temperature	4.6.8.1	1	X	1	X
Low Temperature	4.6.8.2	2	X	2	X
Temperature Shock	4.6.8.3	3	X	3	X
Elevated Temperature	4.6.8.4	4	X	4	X
Dust	4.6.8.5	5	X	5	X
Frozen Sand	4.6.8.6	6	X	6	X
Salt Fog	4.6.8.7	7	X	7	X
Altitude	4.6.8.8	8	X	8	X
Vibration	4.6.8.9	X	1	X	1
Shock	4.6.8.10	X	2	X	2
Humidity	4.6.8.11	X	3	9	3
Static Load	4.6.9	X	4	10	4
Umbilical Hook Support Assembly	4.6.10	X	5	X	5
Firing Test	4.6.11	9	X	X	6

1. Test sequence is denoted by the numbers in the sample columns. An X denotes test not required.
2. A contractor performing individual tests *prior to submission for acceptance* under the contract or purchase order is not required to perform the insulation resistance test at high temperature or the nitrogen gas leakage test at low temperature.
3. The electromagnetic compatibility test shall not be conducted by the contractor. The test shall be conducted by the government (procuring activity) approved test laboratory on one unit only.
4. The procuring activity shall decide which tests of Table III, Group 2, each additional launcher in excess of four shall receive.

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4.5.2 **Sampling tests.** A random sampling of launchers shall be selected by the procuring activity from each inspection lot in accordance with Table IV and shall be subjected to the tests listed in Table III, Group 1 and Table III, Group 2. Group 1 tests shall be conducted prior to Group 2 tests. Failure to pass these tests shall be cause for rejection of the lot. The procuring activity reserves the right to delete tests and alter the test sequence of Table III. In the event any sample representing an inspection lot fails to meet the inspection and test requirements outlined herein, the contractor, at the discretion of the procuring activity, may have the opportunity or reworking the entire affected lot. The procuring activity shall determine which inspections and tests of this specification are to be performed on the reworked inspection lot samples. The number of test samples shall be at the discretion of the procuring activity.

TABLE IV

SAMPLING TEST ITEMS

LOT SIZE	NUMBER OF SAMPLES TO BE TESTED
100 or less	2
101 to 300	3
301 to 500	4
501 or more	4 plus 1 for each additional 500 or fraction thereof.

4.6 **Inspection Methods.**

4.6.1 **Inspection-launcher.** The launcher shall be inspected as follows:

4.6.1.1 **Individual tests-visual.** The launcher shall be inspected to insure that it has been properly assembled and adjusted, that the electrical connections are secure and that the workmanship is as specified herein.

4.6.1.2 **First article and quality conformance tests.**

a. The launcher shall be thoroughly examined as authorized by the procuring activity to determine conformance to all of the requirements of the drawings listed on the data list and this specification for the launcher being procured (see 2.1).

b. During quality conformance inspections only, interchangeability of parts shall be demonstrated by the government approved test facility at the discretion of the procuring activity. Parts shall be interchanged between the launcher samples, before or after completion of other inspections or tests, at the discretion of the procuring activity. This demonstration shall only be performed on the power supply, mechanism assembly, nitrogen receiver and the nitrogen receiver's components and only on those component parts of the launcher which have not experienced visible permanent deformation as a result of prior tests. There shall be no selective or hand fitting of component parts.

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c. The launcher shall also be examined to ascertain that the preparation for delivery conforms to contract requirements (see 5).

4.6.2 Nitrogen gas system. The launcher shall be subjected to the following tests:

CAUTION

Do not pressurize nitrogen receiver assemblies at high fill rates. Filler valves and receiver valves fitted with integral or poppet-type valve seats may become damaged due to overheating. Do not lubricate rubber or plastic seals (O-rings, valve seats, etc.), threads or other mating parts that might, thereby contaminate the gas system.

4.6.2.1 Cleanliness. The receiver assembly submitted with the launcher shall be pressurized through the charging valve to 2000 ± 50 psig with nitrogen gas conforming to 3.6.5. Initial background contamination levels of the test gas shall be recorded. After 48 ± 1 hours, the gas samples shall be collected from the forward connector, J2, of the power supply or dummy power supply, P/N 58A164D710. A minimum of four (4) samples shall be taken. The final gas sample withdrawn from the nitrogen receiver must be utilized for particle examination. Remaining samples shall be used for evaluating non-particulate contaminants. Unless otherwise specified, all tests shall be conducted with the longitudinal axis of the receiver assembly in a horizontal plane. The gas samples shall conform to the cleanliness requirements of Table I.

4.6.2.2 Leakage. The nitrogen gas system shall be subjected to the leakage test specified herein. All checks shall be conducted with the receiver assembly (pressurized to 3,000 ± 100 psig) and a power supply installed within the launcher. All measurements shall be taken at the gas line at the forward connector (J2) of the power supply and with the control valve solenoid energized. A thermocouple shall be located on the surface adjacent to the entry ports on the aft end of the receiver assembly. Pressure corrections due to gas temperature variance shall be accounted for by utilizing the following variation of the gas law $P_3 = P_2(T_1/T_2)$ where:

P_1 = initial pressure (absolute)

P_2 = pressure at test recording (absolute)

P_3 = pressure at test recording corrected to initial temperature (absolute)

T_1 = initial temperature (Kelvin or Rankin scale)

T_2 = temperature at test recording (Kelvin or Rankin scale)

The leakage pressure shall be considered to be the difference between the pressure at test recording corrected to the initial temperature (P_3) and the initial pressure (P_1).

4.6.2.2.1 Room temperature. The receiver assembly shall be stabilized at room temperature (see 4.3) and the pressure shall be 3000 ± 100 psig. The pressure drop over the next four hours after stabilization shall not exceed 150 psig. Final pressure shall be corrected for temperature variations as noted above. If any leakage has occurred in excess of the allowable limits, additional checks shall be conducted, utilizing a suitable detection device to identify the leakage source.

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4.6.2.2.2 Low temperature. The receiver assembly shall be stabilized at room temperature (see 4.3) and the pressure shall be 3000 ± 100 psig. The ambient temperature shall be reduced to -54°C (-65°F) and held for a minimum of eight hours. After the exposure period, the pressure in the receiver assembly shall be recorded. The pressure drop over the next four hours after the exposure period shall not exceed 300 psig. Final pressure shall be corrected for temperature variations as noted above. If any leakage has occurred in excess of the allowable limits, additional checks shall be conducted, utilizing a suitable detection device to identify the leakage source.

4.6.2.3 Flow rate. Perform a nitrogen gas flow rate test at room temperature. The test shall be conducted with the nitrogen receiver pressurized to 3000 ± 500 psig and a power supply installed within the launcher. The gas system shall deliver a minimum of 25 standard liters per minute. All measurements shall be taken at the forward connector of the power supply, J2, and with the control valve solenoid energized.

NOTE

The gas flow impedance of the missile component shall be simulated by attaching a fitting at the J2 connector which contains an 0.010 inch diameter orifice.

4.6.3 Functional test. The launcher shall be subjected to the following mechanical and electrical tests.

4.6.3.1 Mechanical test. The test shall consist of the following functional checks of the mechanical components to demonstrate that they operate freely without sticking or binding.

a.	Forward Snubbers	(see 4.6.4.1)
b	Aft Snubbers	(see 4.6.4.2)
c.	Detent Lift Mechanism	(see 4.6.4.3)
d.	Striker Points	(see 4.6.4.4)
e.	Launcher Rail	(see 4.6.4.5)
f.	Detent Tests	(see 4.6.6)
g.	Umbilical Hook Support Assembly	(see 4.6.7.1)
h.	Fairings	(see 4.6.7.2)
i.	Nitrogen Gas Receiver- Retention	(see 4.6.7.3.2)
j.	Access Door	(see 4.6.7.4)
k.	Housing	(see 4.6.7.5)

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4.6.3.2 Electrical test. The test shall consist of the following to demonstrate the functional capability of all electrical components and circuits in accordance with the following paragraphs.

a.	Electrical Ground Resistance	(see 4.6.5.1)
b.	Continuity	(see 4.6.5.2)
c.	Safety	(see 4.6.5.3)
d.	Solenoid Operation	(see 4.6.5.4)

4.6.4 Missile accommodation. The launcher shall be subjected to the tests listed in the following paragraphs. During the initial inspection of the launcher, these tests shall be conducted using the Sidewinder 1A and 1C missiles in both the minimum and maximum forward hanger conditions (see 4.3). Thereafter, when required after environmental tests, this test shall be conducted using only a Sidewinder 1C in the maximum forward hanger condition.

4.6.4.1 Forward snubbers. Remove the Detent Hold-Down Pin, MS17984C443, and place the snubbers in the down position. Insert the detent wrench/safety pin and rotate. The snubbers shall move forward without binding. Manually move each snubber individually to the up position and rotate the detent wrench/safety pin. The snubbers shall move to the down position and the clearance between snubbers and launcher rail shall fall between 0.150 inch and 0.190 inch as shown in Figure 3. The snubbers shall be checked with a maximum and minimum forward snubber gage shown in Figure 4. The snubbers shall engage the missile sufficiently to prevent excessive movement.

4.6.4.2 Aft snubbers. Release the aft fairing handle latch and swing the handle down. The snubbers shall move in. Rotate the handle back to the latched position. The snubbers shall move to the full out position and shall not bind. The snubbers shall be checked with a maximum and a minimum aft snubber gage shown in Figure 5. No vertical movement of snubber gage shall be permitted. The snubbers shall engage the missile sufficiently to prevent excessive movement.

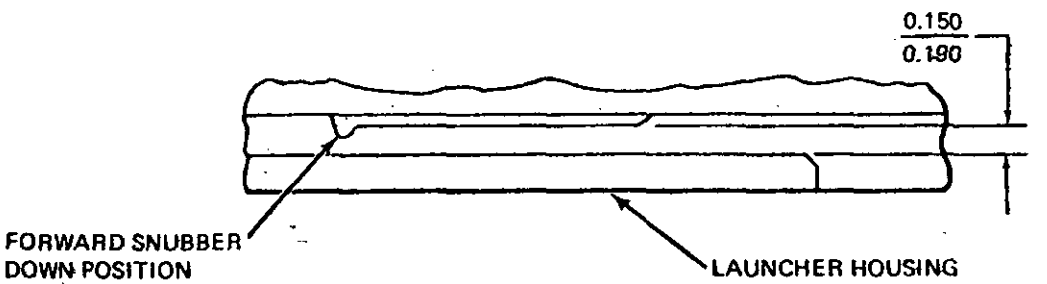
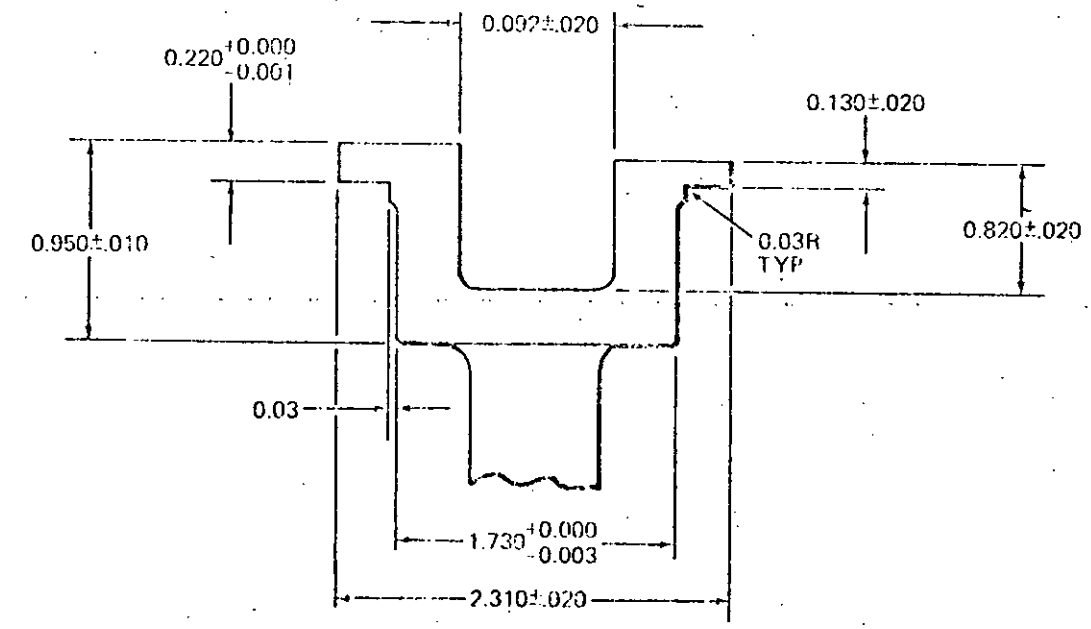
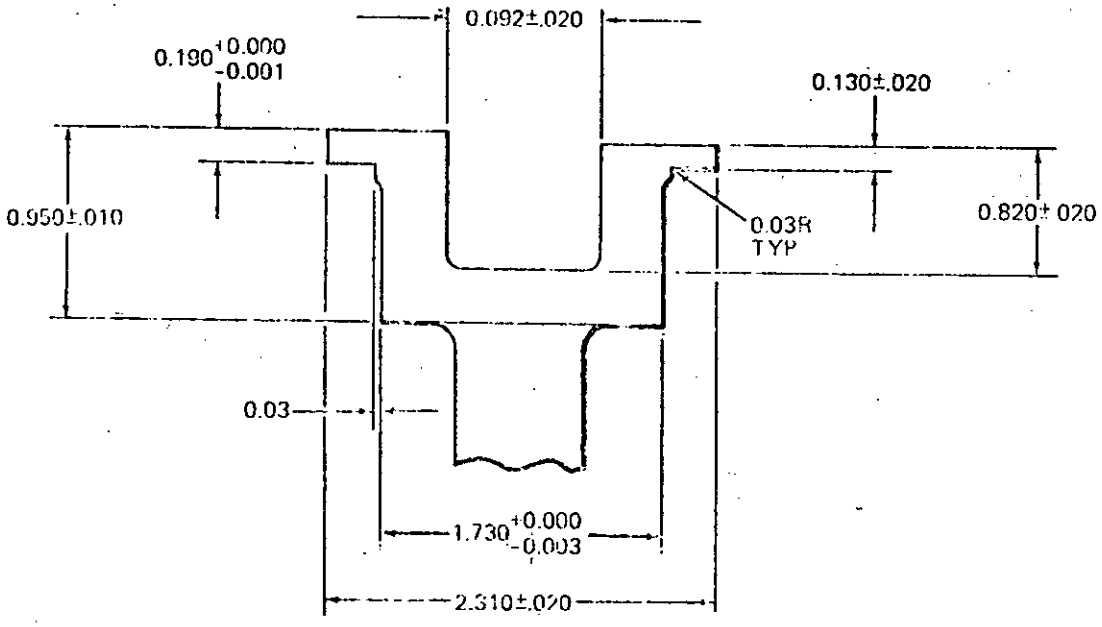


Figure 3. Snubber-to-Rail-Clearance

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MAXIMUM FORWARD SNUBBER GAGE - REFERENCE DRAWING 64A17C101
(NOT TO SCALE)



MINIMUM FORWARD SNUBBER GAGE - REFERENCE DRAWING 64A17C71
(NOT TO SCALE)

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Figure 4. Sidewinder Maximum and Minimum Forward Hanger Gages

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4.6.4.3 **Detent lift mechanism.** The detent lift mechanism shall be subjected to the following:

a. The Detent Hold-Down Pin, MS17984C443, shall be removed, and using the detent wrench/safety pin, the detent shall be manually raised and lowered. The use of additional leverage is not permitted. There shall be no evidence of sticking or binding in either the detent mechanism or snubbers.

b. The detent hold-down pin shall be installed and the detent manually raised until the detent contacts the detent hold-down pin. The aft leg of the detent shall be 0.073 inch minimum below the rail as shown in Figure 6.

c. The detent wrench/safety pin shall be rotated to the maximum release position with the detent hold-down pin removed. The aft end of the detent shall clear the forward hanger.

4.6.4.4 **Striker points.** Striker points shall meet the 0.073-inch minimum dimension as shown in Figure 6.

4.6.4.5 **Launcher rail.** The missile hangers shall not bind in the launcher track or rail.

4.6.5 **Electrical system.** The electrical system of the launcher shall be subjected to the following tests at room temperature with a test missile (see 4.3) installed unless otherwise noted.

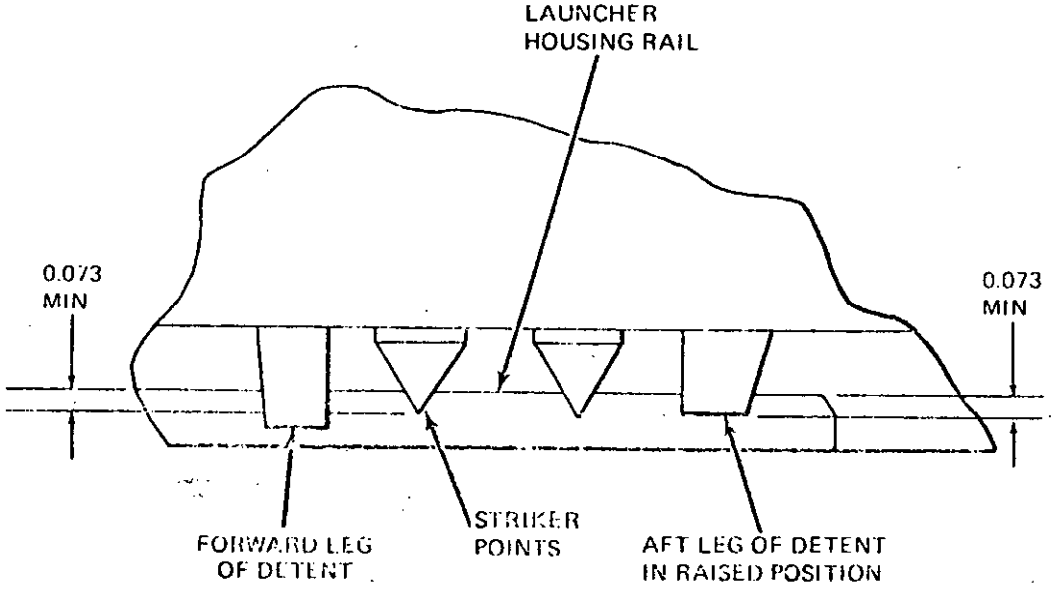
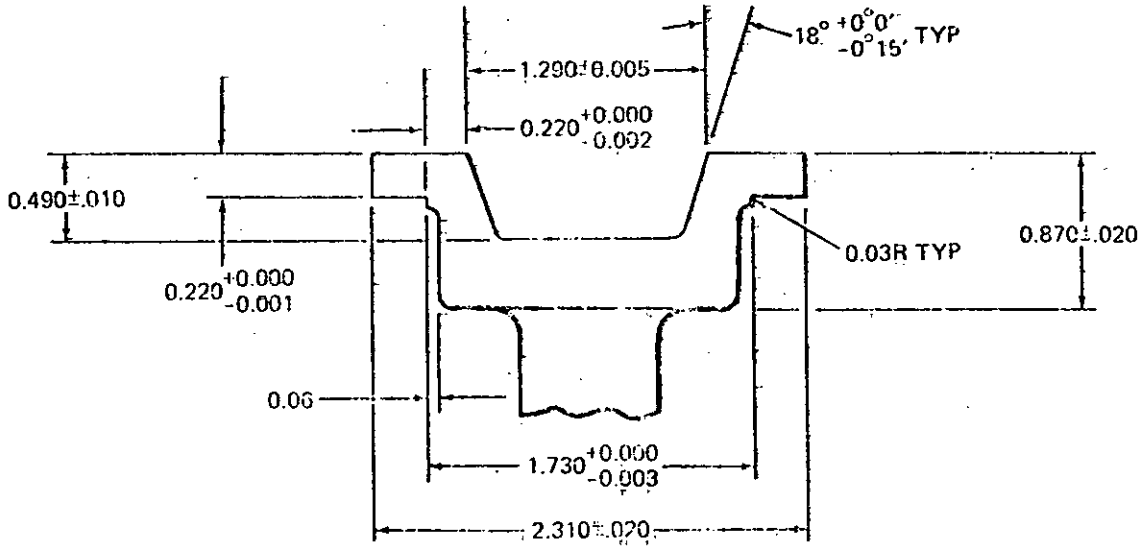
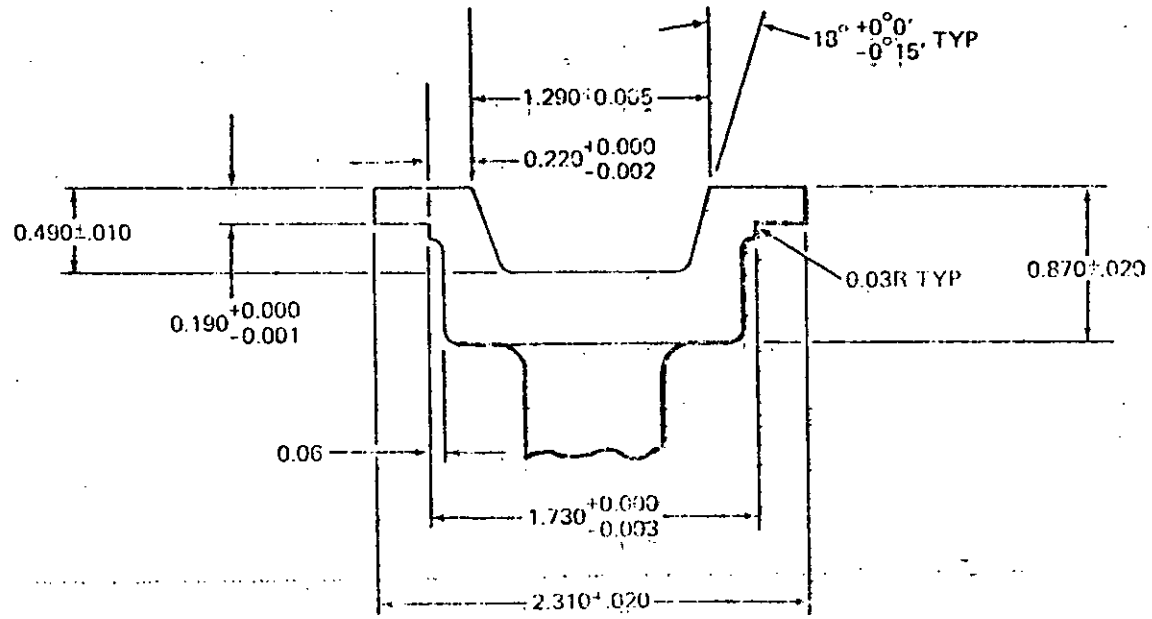


Figure 6. Striker Point and Detent Dimensions

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MAXIMUM AFT SNUBBER GAGE - REFERENCE DRAWING 64A17C72
(NOT TO SCALE)



MINIMUM AFT SNUBBER GAGE - REFERENCE DRAWING 64A17C104
(NOT TO SCALE)

2/626D.6

Figure 5. Sidewinder Maximum and Minimum Aft Hanger Gages

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4.6.5.1 **Electrical ground resistance.** The electrical ground resistance from Pin 16 or Pin 18 of the launcher receptacle J1 to the forward hanger shall be measured. The ground resistance shall not exceed 0.2 ohms.

4.6.5.2 **Continuity.** All electrical circuits shall be checked for continuity as follows:

4.6.5.2.1 **Type I launcher.** The electrical circuit continuity shall be in accordance with Drawing 58A164D571 and PP-2581/A Power Supply Schematic 60A89F125.

4.6.5.2.2 **Type II launcher.** The electrical circuit continuity shall be in accordance with Drawing 64A27D3 and PP-2581/A Power Supply Schematic 60A89F125.

4.6.5.3 **Safety.** The launcher firing and fuze circuits shall be checked as follows.

4.6.5.3.1 **Type I launcher.** The Safety Switches, P/N 58A164C525, shall be in normal-open position with the detent solenoid de-energized and the detent wrench/safety pin out. The launcher firing and fuze circuits at the electrical contact striker points shall show open circuits when a 0.300 ± 0.002-inch diameter round pin is inserted in the detent wrench/safety pin opening and with the detent solenoid energized. The use of a 0.280 ± 0.002-inch diameter pin shall allow the circuits to close, when the detent solenoid is energized.

4.6.5.3.2 **Type II launcher.** The Safety Switches, P/N 58A164C525, shall be in the normal-open position with the detent solenoid de-energized and the detent wrench/safety pin out. The launcher firing and fuze circuit at the aft contact striker point and the target rocket firing circuit shall show open circuits when a 0.300 ± 0.002-inch diameter round pin is inserted in the detent wrench/safety pin opening and with the detent solenoid energized. The use of a 0.280 ± 0.002-inch diameter pin shall allow the circuit to close when the detent solenoid is energized.

4.6.5.4 **Solenoid operation.** The operating voltages of the detent solenoid and control valve solenoid shall be checked as follows:

a. Detent Solenoid - (P/N 58A164D519) 18 to 28 VDC. The current drain shall not exceed 3.5 amp during pull in, and 0.35 amp during hold.

b. Control Valve Solenoid - (P/N 58A164D121) 17 to 29 VDC. The current drain shall not exceed 0.8 amp.

4.6.5.5 **Dielectric breakdown to ground test.** The test shall be conducted on the Detent Solenoid, P/N 58A164D519, and the Control Valve Solenoid, P/N 58A164D121, in accordance with MIL-T-7743.

4.6.5.6 **Insulation resistance test.** The test shall be conducted in accordance with MIL-T-7743 at the temperature condition specified. The power supply shall not be installed.

a. Room temperature

b. High temperature corresponding to Class A insulation.

4.6.5.6.1 **Type I launcher.** The checks shall be conducted on Mechanism Assembly, P/N 58A164D571, as follows:

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NOTE

Ground lead wire (#38) to be broken at terminal E7 (see Drawing 58A164D571).

a. Between each pin of Connector P1, Drawing 58A164C520, and every other pin and ground with the exception of pins 7, 8, 9, 16, 24 and 31.

b. Between the forward striker point contact and ground.

c. Between the aft striker point contact and ground.

4.6.5.6.2 Type II launchers. The checks shall be conducted on Mechanism Assembly, P/N 64A27D3 as follows:

NOTE

Ground lead wire (#38) to be broken at terminal E7 (see Drawing 64A27D3).

a. Between each pin of Connector P1 Drawing 58A164C520, and every other pin and ground with the exception of pins 7, 8, 9, 16, 24 and 31.

b. Between the HVAR connector firing pin and ground.

c. Between the aft striker point contact and ground.

4.6.6 Detent tests. The detents of the launchers shall be subjected to the following checks at room temperature:

4.6.6.1 Detent operational check. The checks shall be conducted as follows:

a. The detent shall remain locked when the detent solenoid is energized with the combination detent wrench/safety pin installed.

b. The detent shall remain locked with the detent solenoid de-energized and the detent wrench/safety pin out.

c. The detent shall unlock with the detent wrench/safety pin out and the detent hold-down pin in. The solenoid shall be energized with 18 VDC.

4.6.6.2 Detent locked. A 2,200 pound force shall be applied to the longitudinal axis of the test missile in both the forward and aft direction. The Detent Hold-Down Pin, MS1798-1C443, and detent wrench/safety pin should be removed and the detent solenoid de-energized. During the initial mechanical inspection of the launcher, this test shall be conducted using the Sidewinder 1A and 1C missiles in both the minimum and maximum hanger conditions (see 4.3). Thereafter, when required after environmental tests, this test shall be conducted using only a Sidewinder 1C in the maximum forward hanger condition (see 4.3). The detent shall not release nor sustain any damage as a result of this test.

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4.6.6.3 **Detent unlocked**. A forward force shall be applied to the longitudinal axis of the test missile. The detent shall release the test missile at a force of not less than 300 pounds nor greater than 800 pounds and shall not sustain any damage. The detent wrench/safety pin shall be removed and the detent solenoid energized. The Detent Hold-Down Pin, MS17984C443, should not be removed for this test. The test shall be conducted using either a Sidewinder 1A or 1C missile in both the minimum and maximum hanger conditions (see 4.3).

4.6.7 **Mechanical checks**. The launcher shall be subjected to the following checks at room temperature.

4.6.7.1 **Umbilical hook support assembly**. The torque required shall be a minimum of 30 inch-pounds when the umbilical hook support assembly is rotated downward to the stop.

4.6.7.2 **Fairings**. The fairings shall operate freely without the use of supplementary mechanical leverage when the forward and aft fairings are operated by hand.

4.6.7.3 ***Nitrogen gas receiver***. The checks shall be conducted as follows:

4.6.7.3.1 **Installation**. The nitrogen receiver shall screw into the control valve assembly by hand without binding and shall not prevent the aft fairing from closing.

4.6.7.3.2 **Retention**. The retention assembly shall hold the nitrogen receiver firmly in place and restrain the nitrogen receiver from rotating when 150 inch-pounds torque is applied to the aft end of the receiver.

4.6.7.4 **Access doors**. The access doors shall permit unimpaired use of the detent wrench/safety pin and the torque wrench adapter (hexagonal extension).

4.6.7.5 **Housing**. The forward bore of the housing with dimensions as shown in Figure 7 shall accept a power supply conforming to MIL-P-81001.

4.6.8 **Environmental tests**. Unless otherwise noted, the environmental tests shall be conducted on the launcher with the following equipment installed:

a. Nitrogen receiver pressurized to 3000 ± 100 psig.

b. Power supply conforming to 4.3.

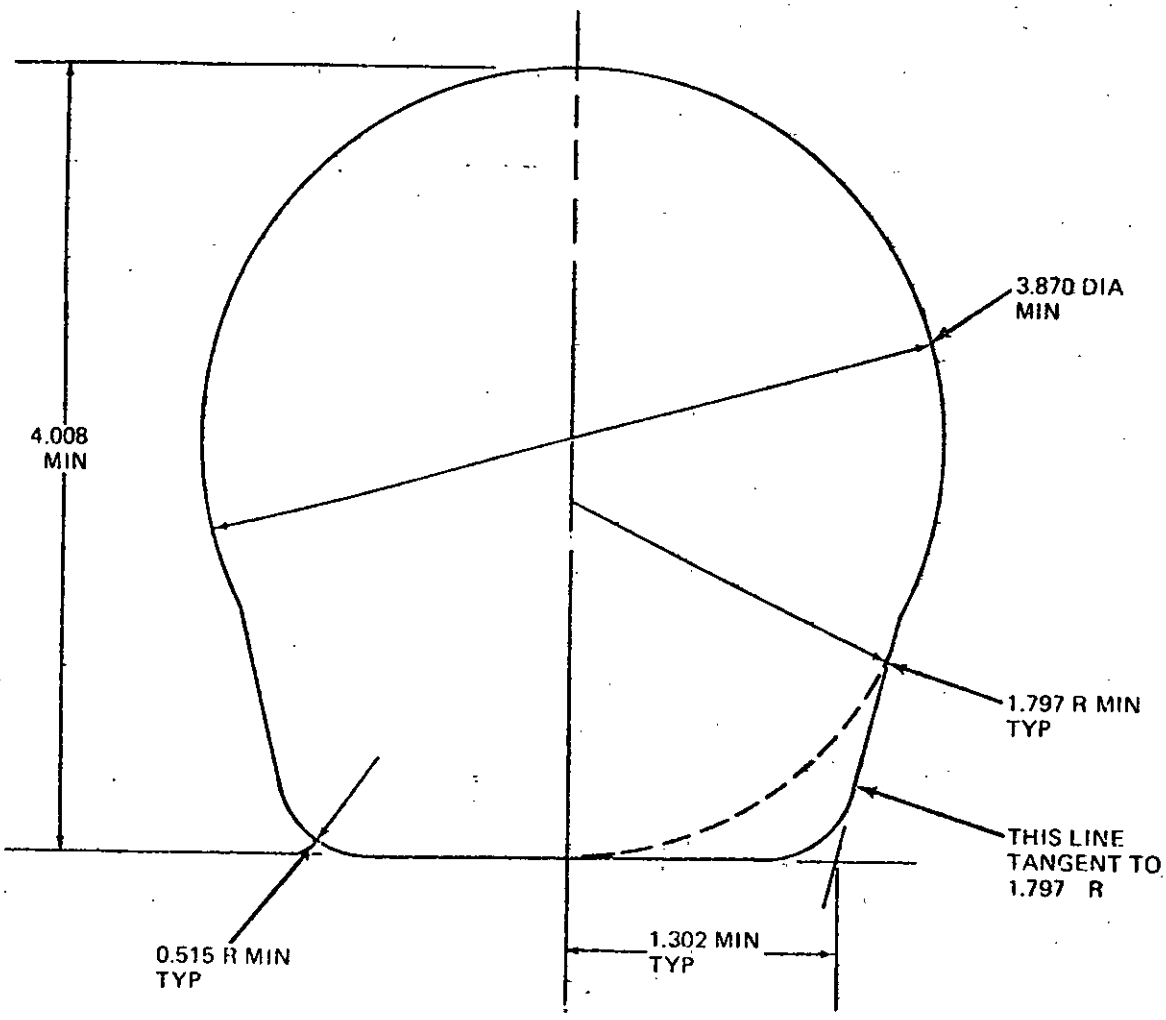
c. Sidewinder 1C test missile with maximum forward hanger conditions (Type I launcher).

d. Sidewinder 1A test missile with maximum forward hanger conditions (Type II launcher).

4.6.8.1 **High temperature**. The launcher shall be subjected to the following tests.

4.6.8.1.1 The outside surface of the entire launcher housing shall be subjected to a temperature of 54°C (130°F) for a period of four hours. The power supply shall be operating as specified in 4.3 and the nitrogen receiver pressurized to 2,500 ± 100 psig at room temperature. During the last one-half hour of this period, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d.

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NOTE: BORE SHALL BE 17.70 INCHES MINIMUM DEPTH

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Figure 7. Dimensions for the Forward Bore of the Housing

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4.6.8.1.2 Within five minutes after completion of the above test, the temperature of the outside surface of launcher housing shall be raised to 110 ± 6°C (230 ± 10°F) and sustained for ten minutes. During the last two minutes, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d.

4.6.8.1.3 Within two minutes after completion of the above test, the temperature of the outside surface of the launcher housing shall be raised to 138 ± 6°C (280 ± 10°F) and sustained for six minutes. During the last two minutes, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d. At the completion of the six-minute period, the launcher housing temperature shall be returned to room temperature in one hour by passing air over the surface of the launcher nose fairing. At the end of the one hour cooling period, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d.

4.6.8.2 Low temperature. The launcher shall be placed within a chamber whose ambient temperature is -54°C (-65°F) for a period of not more than four hours. The power supply shall not be operating and the nitrogen receiver pressurized to 3,000 ± 500 psig at room temperature. During this period, the nitrogen system leakage test shall be conducted as specified in 4.6.2.2. At the completion of the four-hour period, the test chamber ambient temperature shall be reduced to -62°C (-80°F) and maintained at test temperature for a period of 12 hours. The temperature of the test chamber shall then be raised to -54°C (-65°F) and maintained for an additional 8-hour period. During the last one-half hour of the 8-hour period, a replacement nitrogen receiver pressurized to 3,000 ± 500 psig shall be installed in the launcher, the power supply shall be operated (see 4.3) and the launcher tested in accordance with 4.6.2.3 and 4.6.3.2.a., b., d.

4.6.8.3 Temperature shock. After completion of 4.6.8.2 and within ten minutes, the outside surface of the entire launcher housing shall be raised to 138 ± 6°C (280 ± 10°F) and maintained for six minutes. The power supply shall be operating (see 4.3) from start of the temperature increase through the six-minute period. During the last two minutes, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d. At the completion of the six-minute period, the launcher housing temperature shall be returned to room temperature in one hour by passing air over the surface of the launcher nose fairing. At the end of the one-hour cooling period, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d. A visual examination shall be conducted after completion of all tests (see 4.6.8.1, 4.6.8.2 and 4.6,8.3) to determine any deterioration.

4.6.8.4 Elevated temperature. The launcher shall be subjected to the following tests:

4.6.8.4.1 The outside surface of the entire launcher housing shall be subjected to a temperature of 93°C (200°F) for four hours. Only a dummy power supply (see 4.3) shall be used. The nitrogen receiver shall be pressurized to 2,200 ± 100 psig at room temperature. During the last one-half hour of this period, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d.

4.6.8.4.2 At the completion of the above test, and within five minutes, the temperature of the outside surface of the launcher housing shall be raised to 177 ± 6°C (350 ± 10°F) and sustained for fifteen minutes. During the last two minutes, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d. At the completion of the fifteen-minute period, the launcher housing temperature shall be returned to room temperature in one hour by passing air over the surface of the launcher nose fairing. At the end of the one-hour cooling period, the launcher shall be tested in accordance with 4.6.2.3 and 4.6.3.2a., b., d. Additionally, a visual (internal and external) examination shall be conducted after completion of all tests of 4.6.8.4.1 and 4.6.8.4.2 to determine any deterioration.

4.6.8.5 Dust. The launcher shall be subjected to a dust test in accordance with MIL-STD-810, Method 510 Procedure I. The power supply shall not be operating. All connectors normally connected

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shall be protected during the climatic conditioning. At the end of the exposure period, the launcher shall be tested in accordance with 4.6.3.

4.6.8.6 Frozen sand. The launcher shall be subjected to two cycles of a frozen sand test in accordance with MIL-T-7743. The power supply shall not be operating. All connectors normally connected shall be protected during the climatic conditioning. At the end of each freeze, the launcher shall be tested in accordance with 4.6.3.1a., g., i., k., and 4.6.3.2a., b., d.

4.6.8.7 Salt fog. The launcher shall be subjected to the salt fog test in accordance with MIL-STD-810, Method 509, Procedure I. The exposure period shall be 50 hours. The power supply shall not be operational. All connectors normally connected shall be protected during the climatic conditioning. At the completion of testing, the launcher shall be tested in accordance with 4.6.3.

4.6.8.8 Altitude. The launcher shall be subjected to the altitude test in accordance with MIL-STD-810, Method 500, Procedure II. The altitude shall be 80,000 feet. The power supply shall be operating. While at the above conditions, the launcher shall be tested in accordance with 4.6.2.3, 4.6.3.2a., b., d., 4.6.5.1, and 4.6.6.3.

4.6.8.9 Vibration. The launcher shall be subjected to the following test.

4.6.8.9.1 General. The vibration tests specified herein shall be performed in each of three mutually perpendicular axes in the order specified herein. All testing shall be performed in one axis prior to testing in the next axis. As indicated herein, the unit shall be checked for proper operation. Inadvertent release of the item during vibration from any cause shall be considered a failure. Structural damage to the launcher, or damage to any component part, shall be considered a failure.

4.6.8.9.2 Test item operation. The nitrogen receiver shall not be pressurized during this test. Instead, a dummy nitrogen receiver shall be weighted to simulate the weight and center of gravity of a nitrogen receiver pressurized to 3000 psig at room temperature. After the unloaded and loaded sinusoidal and the random unloaded conditions are completed in an axis, the launcher shall be tested with the original nitrogen receiver installed and pressurized, as specified in 4.6.2.3 and 4.6.3.2a., b., c. The random loaded condition will then be completed and the launcher tested as specified in 4.6.3. These operational checks shall be made in this manner in each of the three mutually perpendicular axes.

4.6.8.9.3 Mounting techniques. The launcher shall be attached to the vibration exciter table by means of a rigid fixture capable of transmitting the vibration conditions specified herein. The launcher shall be mounted to the text fixture in the normal (vertical in reference to the aircraft) service configuration. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of undesirable responses in the test setup. Whenever possible, the item weight shall be distributed uniformly on the vibration exciter table in order to minimize effects of unbalanced loads. Vibration amplitudes and frequencies shall be measured by techniques that will not significantly affect launcher input control or response.

The vibration input shall be measured at the following locations:

a. On the fixture, adjacent to the forward launcher mounting bolt.

b. On the fixture, adjacent to the aft mounting bolt.

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c. On the launcher, adjacent to the forward mounting bolt.

d. On the launcher, adjacent to the aft mounting bolt.

The control signal shall be the average of all of the above, unless otherwise specified. The transverse motion of the input monitoring points shall be minimized, and should be limited to 100 percent of the input motion. In addition, response accelerometers shall be placed at the following locations:

a. Triaxial pick-up on the forward end of the launcher housing.

b. Triaxial pick-up on the aft end of the launcher housing.

4.6.8.9.4 Sinusoidal test level and sequence. The launcher shall be subjected to sinusoidal vibration along each of three mutually perpendicular axes. The frequency shall be ± 2 percent and the amplitude shall be ± 5 percent.

a. Sinusoidal search unloaded - Sweep for 30 minutes from 5 to 2000 Hz at one octave per minute in accordance with the following requirements:

Frequency	Displacement or acceleration
5 - 20 Hz	0.100 inch da
20 - 33 Hz	± 2.0 g's
33 - 74 Hz	0.036 inch da
74 - 2000 Hz	± 10.0 g's

b. Sinusoidal search loaded - Sweep for 30 minutes from 5 to 2000 Hz at one octave per minute in accordance with the following requirements:

Frequency	Displacement or acceleration
5 - 20 Hz	0.100 inch da
20 - 33 Hz	± 2.0 g's
33 - 52 Hz	0.036 inch da
52 - 2000 Hz	± 5.0 g's

4.6.8.9.5 Random test level and sequence. The launcher shall be subjected to a random vibration along each of three mutually perpendicular axes. The instantaneous random vibration acceleration peaks shall be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than + 100, -50 percent (± 3 dB) between 20 Hz and 2,000 Hz, except that deviations as large as + 300, -75 percent (±6 dB) shall be allowed over a cumulative bandwidth of 100 Hz, maximum, between 500 and 2,000 Hz.

Tolerance levels in terms of dB are defined as:

$$dB = 10 \log \frac{W_1}{W_0}$$

Where W_1 = measured acceleration power spectral density in g^2/Hz units.

The term W_0 defines the specified level in g^2/Hz units.

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Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT = 50, minimum. Specific analyzer characteristics shall be as specified below or equivalent.

a. On-line, contiguous filter, equalization/analysis system having a bandwidth = B = 50 Hz maximum.

b. Swept frequency analysis systems characterized as follows:

1. Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B = 20 Hz, maximum between 20 and 200 Hz.

B = 50 Hz, maximum between 200 and 2,000 Hz.

(b) Analyzer averaging time = T = 2RC = 1 second, minimum, where T = true averaging time and RC = analyzer time constant.

(c) Analysis sweep rate (linear) = $S = \frac{B}{4RC}$ or $\frac{B^2}{8}$ Hz/sec, whichever is smaller.

2. Constant percentage bandwidth analyzer.

(a) Filter bandwidth = pf_c = one third octave minimum (0.23f) where p = percentage and f_c = analyzer center frequeney.

(b) Analyzer averaging time = $T = \frac{50}{pf_c}$ minimum.

(c) Analysis sweep rate (logarithmic) = $S = \frac{pf_c}{4RC}$ or $\frac{(pf_c^2)}{8}$ Hz/second, whichever is smaller.

c. Digital power spectral density analysis systems employing quantization techniques shall provide accuracies corresponding to the above approach.

The random vibration test levels shall be as follows:

1. Random vibration unloaded. Conduct for 40 minutes in each of the three mutually perpendicular axes. The test levels shall be a flat spectrum in the range of 20 to 2,000 Hz at a PSD level of 0.02 g^2Hz with an overall test level of 6.3 g's rms. The analysis shall be as follows:

Bandwidth	Analysis filter
20 - 200 Hz	20 Hz filter
200 - 2000 Hz	50 Hz filter

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The analysis shall yield a statistical accuracy of 120 degrees of freedom.

2. Random vibration loaded. Conduct for 40 minutes in each of the three mutually perpendicular axes. The test levels shall be a flat spectrum in the range of 20 to 2,000 Hz at a PSD level of 0.018 g^2Hz with an overall test level of 5.9 g's rms. The analysis shall be as follows:

Bandwidth	Analysis filter
20 - 200 Hz	20 Hz filter
200 - 2000 Hz	50 Hz filter

The analysis shall yield a statistical accuracy of 120 degrees of freedom.

4.6.8.10 Shock. The launcher shall be subjected to the following procedure.

4.6.8.10.1 General. Unless otherwise specified, the launcher shall be subjected to the required acceleration in both directions along the three principal axes of the launcher. The peak applied values of the accelerations shall be within plus 15, minus 0 percent of the values specified in the following and shall reach peak value in not more than 0.60 second and not less than 0.015 second. The waveform of acceleration shall resemble a half sine wave. As indicated herein, the launcher shall be checked for proper operation. Inadvertent release of the missile during shock from any cause shall be considered a failure. Structural damage to the launcher or to any component part, shall be considered a failure.

4.6.8.10.2 Test item operation. The nitrogen receiver shall not be pressurized during this test. Instead, a dummy nitrogen receiver shall be weighted to simulate the weight and center of gravity of nitrogen gas at 3000 psig and at room temperature. After the loaded conditions are completed in each direction of each axis, the launcher shall be tested as specified in 4.6.3.

4.6.8.10.3 Mounting techniques. The launcher shall be mounted on an Aero 3A Pylon, or equivalent. The shock inputs shall be located at the center of gravity of the launcher for all the unloaded conditions and the center of gravity of the missile for all the loaded conditions.

4.6.8.10.4 No load. In order to check for no-load dynamic effects and inadvertent release, the launcher shall be subjected to an acceleration of 25 g's along all the principal axes while supporting no load.

4.6.8.10.5 Normal load. In order to check for dynamic effects and inadvertent release, the launcher shall be subjected to two accelerations in accordance with Table V along all the principal axes while supporting a test missile.

4.6.8.10.6 Catapult and arrested landing. In order to check for dynamic effects and inadvertent release for catapult and arrested landings, the launcher shall be shocked in the longitudinal forward and aft directions, to the levels specified in Table V. The acceleration shall reach peak value in not more than 0.75 second and not less than 0.4 second.

4.6.8.11 Humidity. The launcher shall be subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure I, except that four 24-hour cycles shall be substituted for ten 24-hour cycles. At the completion of the humidity tests, the launcher shall be tested as specified in 4.6.3.

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TABLE V

APPLIED SHOCK LOADS

DIRECTION	APPLIED SHOCK LOAD (g's)
Vertical Up	7.1
Vertical Down	12.5
Longitudinal Forward	9.8
Longitudinal Aft	9.8
Transverse Port	8.2
Transverse Starboard	8.2

4.6.9 Static load. The launcher shall be subjected to the conditions specified in Table VI. The loads shall be applied on the longitudinal centerline of the test missile at its center of gravity. After reaching the specified load, the load shall be reduced to zero and the test missile manually disengaged from the launcher. Launcher function shall not be degraded nor shall there by any permanent deformation or damage noticeable by visual examination after return to zero load. The launcher shall not release the test missile while at or below the specified loads. At the completion of all the conditions of Table VI, the launcher shall be tested as specified in 4.6.3.

4.6.10 Umbilical hook support assembly. The launcher shall be subjected to a forward axial load of 1,000 pounds, applied to the umbilical hook support assembly. The load shall be applied with the spring-loaded arm assembly bottomed on the launcher rails and the forward axial load applied parallel to the launcher rails. The umbilical hook support assembly shall withstand the load without permanent deformation or damage noticeable by visual examination after return to zero load.

TABLE VI

APPLIED STATIC LOADS

CONDITION	P_z VERTICAL (LBS.)	P_y SIDE (LBS.)	P_x LONGITUDINAL (LBS.)
1	-2400	6000	-3000
2	-2400	6000	3000

SIGN CONVENTION

$+P_x$ = Longitudinally Aft
$+P_y$ = Side to Port
$+P_z$ = Vertical Up

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4.6.11 Firing. The launcher shall be subjected to 100 simulated firings. After each firing, the launcher shall be visually checked for any signs of degradation. The test fixture shall impart a velocity to the test missile so that it will depart the launcher rail at a velocity of 92.5 ± 7.5 feet per second. The launcher shall be tested as specified in 4.6.3 after 50 and 100 simulated firings.

4.6.12 Electromagnetic compatibility. The launcher shall be subjected to the requirements of MIL-I-6181. During testing, the power input and load conditions of MIL-I-81001 shall apply.

5. PREPARATION FOR DELIVERY

5.1 Launcher preservation and packaging. Preservation and packaging shall be in accordance with Level A or C of MIL-STD-794, as specified in the contract or purchase order.

5.1.1 Level A. Launchers shall be unit packaged, one each, in accordance with Method II submethod IIE of MIL-P-116, without the use of preservation compound.

5.1.2 Level C. Launchers shall be unit packaged, one each, in accordance with Method III, of MIL-P-116, without the use of preservation compound.

5.2 Packing. Shipping containers shall be suitable for Level A or Level C as specified in MIL-STD-794.

5.2.1 Level A. Launchers shall be unit packed in shipping containers conforming to PPP-B-601.

5.2.2 Level C. Launchers shall be unit packed in shipping containers conforming to PPP-B-640.

5.3 Unitized loading. Unitized loading for the launchers is authorized as specified in MIL-STD-794.

5.4 Marking. All shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use.

a. The LAU 7/A Series guided missile launcher when attached to an aircraft provides a self-contained launching system for the Sidewinder 1C missile.

b. The LAU 7A/A Series guided missile launcher when attached to an aircraft provides a self-contained launching system for the Sidewinder 1A missile or the HVAR target rocket (see 6.5).

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6.2 **Information for contracting officer.** Contracts or purchase orders shall specify the following:

a. Title, number, revision letter and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing required (see 5.).

c. Samples subjected to sampling inspection (see 4.4.1 and 4.5.2) shall not be considered or accepted as part of the contract or purchase order.

d. Rework and retest requirements (see 4.5.2) for inspection lot samples.

e. Items of data required (see 6.3).

f. Name and location of government-approved test laboratory.

6.2.1 **Contract or purchase order provision.** Contracts or purchase orders shall specify the following provision for first article inspection. "When a contractor is in continuous production of the launchers applicable to this specification from contract to contract, consideration shall be given to waive the first article inspection. However, waiver of first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract." The words "purchase order" shall be substituted throughout for the word "contract," when applicable.

6.3 **Data.** For the information of the contractors and contracting officers, any of the data specified in (a) the following subparagraphs, (b) applicable documents listed in Section 2 of this specification, or (c) referenced lower tier documents shall be furnished to the government unless otherwise specified in the contract or purchase order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made part of the contract or purchase order. NAVAIR Form 4200/25 (Drawings, Lists and Specifications Required) shall be attached where applicable.

6.3.1 **First article.** When first article samples are submitted to the government (procuring activity) approved test laboratory (see 4.4), the results of the tests shall be presented by a complete inspection report showing the results of the contractor's inspections. The inspection report shall include the following:

a. Report of all inspections graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification and applicable military standard referring specifically to paragraph numbers. When a requirement is considered to be not applicable, the report shall so state.

b. Diagrams of inspection setups complete with description of inspection equipment, test and inspection procedures.

c. Reproducible outline and description of inspection conditions. Where inspections specified in this specification are not considered applicable, the reason, and the substituted inspection shall be clearly described.

d. Copies of test and inspection log sheets.

e. Photographs when available.

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6.3.2 **Quality conformance.** When quality conformance inspection lot samples are submitted to the government (procuring activity) approved test laboratory (see 4.5), the results of the tests shall be presented by a complete inspection report showing the results of the contractor's inspection. The inspection report shall include the following:

a. Report of all inspections graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all requirements of this specification and applicable military standard referring specifically to paragraph numbers. When a requirement is considered to be not applicable, the report shall so state.

b. Diagrams of inspection setups complete with description of inspection equipment, test and inspection procedures.

c. Reproducible outline and description of inspection conditions. Where inspections specified in this specification are not considered applicable, the reason, and the substituted inspection shall be clearly described.

d. Copies of test and inspection log sheets.

e. Photographs when available.

6.4 **Quality assurance definitions.** Definitions for quality assurance terms are in accordance with MIL-STD-109.

6.5 The Target Rocket TDU-11 is referred to herein as "Target Rocket" or "HVAR" (Hyper Velocity Aerial Rocket).

6.6 **Availability of dummy stores for testing.** The contractor shall submit requests to the procuring activity for dummy and inert stores to be used for testing as required by this specification

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